Title: Capstone Project Proposal: NLP and use of AI in generating human-like response

Capstone Project Proposal Submitted in Partial Fulfillment of the Requirements for the Master of Science in Data Science, Meharry Medical College

by

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Submission Date: December 7 2025

Expected Completion Date: April, 2026

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1 Project Description

I propose to build a chatbot that will be integrated on the SAC's website. The purpose is to guide prospective students through the admission process to the SACS program. Showcases the SACS program uniqueness in computational science with health equity and biomedical research and provide support outside business hours.

1.1 Problem Statement

The capstone proposed involves building a chatbot that can supplement a traditional website page making it easy and accessible to navigate and find information (For example, using a tertiary institution). The problem commonly encountered involves dealing with enormous amount of academic application which can be a bit tedious on the institutions.

1.2 Objective Statement

The goal of the project is to adequately improve user interaction and navigation on web-pages by using machine learning algorithm to help provide quick information saving ample of time that can be devoted to other resources. This saves cost by reducing intermediary or agents that often provide redundant information due to the complexity of navigating web-pages with vast amount of information. The information gathered can be funneled to guidance counselors for proper evaluation. This reduces the burden for an admission team

1.3 Proposed Solution

The solution to the problem proposed involves using natural language processing to create a chatbot that can be easily integrated into a website. This provide feedback to users and relay information back to the admission team simplifying the process of admission such as checking eligibility and providing direct links to resources. This saves time for both users and administrative officers.

2 Literature Review

My project does not aim to fill any gaps in the literature but proposes to implement the latest up to date research on chatbot and apply it to a novel webpage (SACS webpage) with the hope of highlighting an overall understanding of course material (Natural language processing, Data management, machine learning ... etc.) completed throughout the graduate program. My project aims to paint an understanding on the latest developments in human-computer interaction by applying this concept to my capstone project.

Previous research on my proposed topic show that chatbots has been around for quite a while, the first know text-based chatbot ELIZA was developed at MIT by Joseph Weiznenbaum (a German-American computer scientist) in the 1960s which stimulated human-like conversation, this was an attempt to beat the Turing test (Sowjanya et al., 2024), this chatbot acted as a psychotherapist. PARRY was also developed by Kenneth Colby (an american psychiatrist) at Stanford around 1968

- 1972, similar to ELIZA, it was a rule based chatbot designed to mimic a person with paranoid schizophrenia (Alnefaie et al., 2021). The first speech-based dialoged systems (Jabberwacky) developed by Rollo carpenter (a british computer scientist) used voice-controlled user interfaces developed in 1988 and was capable of beating the turing test, it was designed to mimic human interaction and carry out conversation with users. Today's chatbot uses artificial intelligence for conversational dialogue.

There are different types of chatbots which are listed as follow:

- Rule-based chatbot
- AI – powered chatbots
- Voice chatbots
- Generative AI chatbots
- Hybrid chatbots

2.1 Rule based chatbots

Rule based chatbots are based on set of pre-defined rules often using a decision tree algorithm (conditional statement). Chatbots can be directed to response based on the previous input and can be negated (opposite). This chatbots are easy to train well and are known to have been around for decades (prior to the advent of the internet such as the ELIZA chatbot). Rule-based chatbots answer users' queries by looking for patterns matches; hence, they are likely to produce inaccurate answers when they come across a sentence that does not contain any known pattern (Finn et.al 2025) queries. Due to the inability to perform conversation design for every possible query, the limited rule-based chatbots often get stuck when they can't understand the user's request (Finn et.al 2025).

2.2 AI chatbots

AI-powered chatbots are based on machine learning algorithms that allow them to learn from an existing database of human conversations. In order to do so, they need to be trained through Machine Learning algorithms that can train the model using a training dataset. Advantages of AI chatbots are they understand questions no matter how they are phrased. They often use NLP; natural language processing to analyze information and text data to create a response output. (Finn et.al 2025) With Deep learning, the longer an AI chatbots operates, the better it can understand user goals and provide more detailed, accurate responses, as compared to a chatbots with recently integrated algorithm-based knowledge. Conversation AI chatbots can remember conversations with users and incorporate this context into their interactions (Finn et.al 2025)

2.3 Generative AI chatbots

Generative AI chatbots are another subset of AI powered chatbots that use large language modelling (LLM). These models are thus able to create entirely new sentences to respond to users' queries; Geneerative AI are fluent in understanding common language, can adapt to a user's style of conversation and use empathy when answering users' questions (Finn et.al 2025). While conversational AI chatbots can digest a user's questions or comments and generate a human-like response, generative AI chatbots can go a step further by generating new content as the output. (Think about generating a new image or video of a human)

2.4 Voice chatbots

Voice chatbots allow users interact with the bot it speak to. Training can be based using rule-based, LLM, or NLP. Chat and voice bots both aim to identify user needs and provide helpful responses. However, voice chatbots can offer a quicker and more convenient communication method, as it's easier to get a real-time answer without typing or clicking through drop-down menu options. Training and type of algorithm is crucial in understanding the complexity of chatbots. Recommended to use advanced algorithm when training voice chatbots (Finn et.al 2025).

2.5 Hybrid chatbots

A hybrid chatbot is a conversational AI system that combines rule-based logic with machine learning capabilities. The combination of both can deliver a versatile user experience that handles a range of tasks varying in difficulty due to the integration of AI technology (Finn et.al 2025). Rule-based chatbots function on a set of predefined rules and scripts and therefore provide structure, while AI has learning potential for more complex interactions. The hybrid chatbot provides the best of both systems in one single system and delivers a vast user experience that is straightforward and personalized (Finn et.al 2025)

3 Methodology and Approach

The major activities that will be taken to accomplish my goal can be divided into various segments. First, an appropriate target will be chosen. For my project, I have decided to focus on admissions into the Meharry SACs program. The type of chatbot framework will be considered. My scope will be narrowed between Large language model (LLM chatbot) or traditional NLP bot. Next, a platform will be utilized. With the permission of the department, I aim to use the SACS website page. The tech stack framework will also be analyzed. This will be divided into a backend to process and store data, a framework, a LLM provider and a front-end (to produce output and for interaction). The data and knowledge will be prepared using a core logic such as flow bot (which analyzes previous response and interaction before suggesting or moving to the next phase) or an LLM based bot (that uses system prompt, conversational history, optional tools and optional retrieval to interact with users).

For creating the chatbot. I will be using an LLM to handle conversation between user and the chat bot and a vector database for retrieval. An example of a conversation will be

- "What is the MSDS/PHD program?
- "How much is Tuition, Where can i get financial aid?"

- “What does SACS stand for?”
- “Do I need the GRE?”
- “What is the admission requirement?”

3.1 Details of the Plan

Concretely, my project will use an LLM (via Python) to handle natural conversation. Use retrieval (RAG) over the SACS website pages (scraped or exported) and LLM (GPT) which allow for users to interact with the bot regarding content on the SACS website and store on (vector database). The architecture I plan to use include building and hosting the chatbot outside the SAC’s website and then fully integrating and embedding it on the SAC’s website. The core logic of my project includes using either a flow bot/engine to engage and respond to users based on their previous response or an LLM based bot that uses system prompt, conversational history and optional retrieval. The architecture for the Meharry website chatbot includes:

- Backend / logic done using Langchain with OpenAI and Chroma or FAISS vector store for document retrieval.
- Web user interface (python or react JS) . For python Gradio software.
- Deployment will be done using hugging face spaces or render or any other python app hosting.

Data needed to train the chatbot will be derived from the SAC’s website page and other PDF resources. The data will be cleaned, organized and stored in a vector store.

3.1.1 Research Design and Data Collection Methods

After implementing my chatbot, I will test the applicability by recruiting students and faculty members. First, i will formulate a research question to measure one of the following metrics: usability and experience, usefulness and impact, acceptance/intent to use.

I will then recruit prospective students and faculty members after obtaining consent by sending an invitation to use the newly integrated chatbot. After, interaction a survey will be administered the survey will aim to capture usability and experience, usefulness, intention to use and other metrics

3.1.2 Sampling Techniques and Sampling size

A combination of convenience and purposive sampling techniques will be employed. For the quantitative component, convenience sampling will be used to recruit Meharry SACS students who will voluntarily interact with the SACS chatbot and complete the accompanying online survey. An intended sample size of approximately 20–100 participants is the target to ensure sufficient data for descriptive and correlational analysis while meeting deadlines within the semester timeframe. For the qualitative component, purposive sampling will be used to select 10–30 students from the survey respondents who have interacted with the chatbot and express willingness to participate in follow-up interviews. This approach will enable an in-depth exploration of user experiences, usability perceptions, and areas for improvement. Chatbot usage log data will also be analyzed for all consenting participants to complement self-reported measures with behavioral usage patterns.

3.1.3 Potential limitations of the methodology

I plan to analyze the study's methodology which is subjected to several limitations. First, the use of convenience and voluntary sampling may introduce selection bias, limiting the representativeness of the sample. The relatively small sample size further constrains the generalizability of quantitative findings (the sample may not represent the entire SACS student population. Students who are more comfortable with technology or who are more engaged with campus resources may be overrepresented, while less tech-savvy or disengaged students may be underrepresented). Second, much of the data relies on self-reported perceptions, which may be influenced by social desirability bias. Third, the evaluation is cross-sectional and short-term, capturing initial reactions rather than long-term behavioral outcomes. Additionally, technical constraints of the chatbot and researcher involvement may influence user experiences and interpretation. These limitations are acknowledged and will be addressed through transparent reporting, triangulation of data sources (survey, usage logs, and interviews), and recommendations for future longitudinal research.

4 Deliverables

- Build a chatbot for the SACS webpage
- Run Test
- Survey student and faculty members for pilot testing
- Collect Feedback on user friendliness and usability
- Run qualitative or quantitative analysis on user/'s feedback
- Project Defense and Final project report

5 Technology

- Retrieval Augmented Generation (for retrieving updated infor from the SAC webpage)
- Python (Google Colab) or JavaScript
- LangChain OpenAI (GPT or Gemini)
- ReactJS or Gardio (for frontend user interface)
- FAAIS library (key component of RAG).
- Chroma Vector Database for storing unstructured data (conversation with user)
- Qualtrics or Student Outlook Email for research participation and Feedback

6 Dependencies and Limitations

I propose to create a chatbot that consists of two models, first will be the retrieval of information via RAG (retrieval augmented generation) from the SACS webpage for information that pertains to the user query. This will be achieved via semantic search. Second, a hand off to a LLM (openAI) to process response/output. Since, the conversation and interaction is based on a bimodal system that is dependent on external software. This poses a risk if there is an inherent issue with response processing.

Limitations that I might encounter involves obtaining permission from the institution to conduct a research pilot project. Research participant size can also be a possible limitation due to the class size in the SACS program department.

6.1 Ethical, Privacy and Cyber-security concerns

6.1.1 Privacy

The collection of data poses risk. Building a chatbot involves interaction between a user and the LLM based bot during pre-training, research design and implementation. The research design will attempt to exclude personal de-identifiers like race, gender and age. There are still some risk that include chat logs that could contain sensitive or personal information such as student gpa, email (if redirecting to a human) ... etc. Strict data handling and storage is therefore required to anonymize and store this log securely.

6.1.2 Confidentiality

Likewise, an ethics approval from a governing body is required to handle sensitive information derived from potential students notably during the research process. Confidentiality will be enforced by de-identifying identifiers, asking for consent to partake in research before participation.

6.1.3 Ethics

Ethical rules will be incorporated in the training program to prevent the chatbot from making unfounded assumptions regarding student . Limitations will also be added as a safeguard rail when complex questions are encountered. This limitation will ensure all complex queries are redirected to a human for further analysis.

7 Project Timeline

7.0.1 Table 1: Prospective timeline for accomplishing each task in the capstone project.

Milestone	Task	Expected Completion Date
Background, meeting and capstone framework with supervisor	Provide framework for the capstone project	January 2026
Chatbot Design using LLM and NLP	Design and integrate chatbot on meharry SAC's website	February 2026
Research Design Testing	Test research design on student and faculty	March 2026
Research Design Evaluation	Quantitative and qualitative evaluation and feedback of research design	March 2026
Project Final Report	Written report of final project	April 2026
Capstone Presentation and Project Defense	Present capstone project to faculty and defend capstone project	April 2026

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